Chapter 5: Financial Crisis

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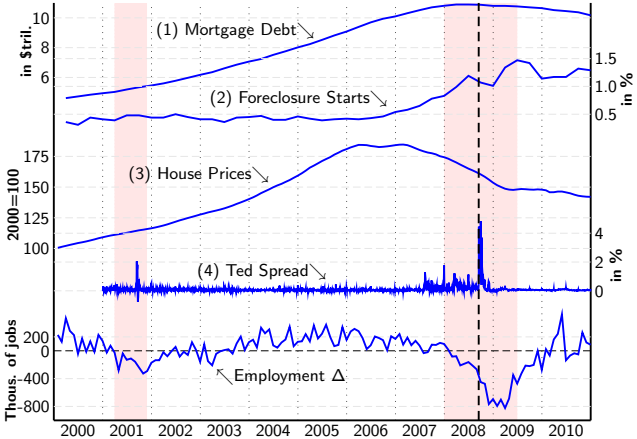
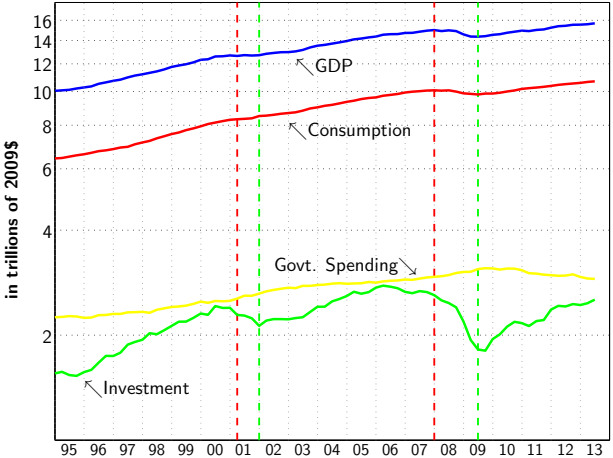
Course Overview



- (1) Debt Growth
- (2) Default
- (3) House Prices
- (4) Financial Crisis
- (5) Institutions
- (6) Expectations

Components of Growth



Expenditure Accounting

- National Income Identity

$$Y_t = C_t + I_t + G_t + NX_t.$$

- Growth:

$$Y_t = \frac{Y_t}{Y_{t-1}} Y_{t-1} = Y_{t-1} + \Delta Y_t \cdot Y_{t-1}$$

- Decomposition.

$$\begin{aligned} Y_t = Y_{t-1} + \Delta Y_t \cdot Y_{t-1} = \\ C_{t-1} + \Delta C_t \cdot C_{t-1} + I_{t-1} + \Delta I_t \cdot I_{t-1} \\ + G_{t-1} + \Delta G_t \cdot G_{t-1} + NX_{t-1} + \Delta NX_t \cdot NX_{t-1}. \quad (1) \end{aligned}$$

- Growth Accounting Equation

$$\Delta Y_t = \Delta C_t \cdot \frac{C_{t-1}}{Y_{t-1}} + \Delta I_t \cdot \frac{I_{t-1}}{Y_{t-1}} + \Delta G_t \cdot \frac{G_{t-1}}{Y_{t-1}} + \Delta NX_t \cdot \frac{NX_{t-1}}{Y_{t-1}}. \qquad (2)$$

The Housing Boom, 2001-2006

From: Q4/2001 To: Q1/2006

Variable	Growth	×Share	=Contribution	Share of Growth
C	14.5	66.9	9.7	66.8
I	30.4	16.7	5.1	35.1
Residential	*35.1*	*5.1*	*1.8*	*12.3*
Non − Res	*28.4*	*11.7*	*3.3*	*22.8*
G	8.4	20.7	1.7	12.0
NX	53.1	-4.1	-2.2	-15.1
Y	14.5	100.0	14.5	100.0

- Consumption accounts for exactly its normal share.
- Strong investment is the key.
- Offset by increased imports (fall in NX)

The Housing Bust, 2006-2009

From: Q1/2006 To: Q2/2008

Variable	Growth	×Share	=Contribution	Share of Growth
C	3.6	66.9	2.4	83.6
I	-10.9	19.1	-2.1	-72.2
Residential	*-40.8*	*6.0*	*-2.4*	*-85.2*
Non − Res	*2.8*	*13.1*	*0.4*	*13.0*
G	4.3	19.6	0.8	29.1
NX	-31.1	-5.5	1.7	59.6
Y	2.9	100.0	2.9	100.0

- Housing bust led to massive fall in residential investment.
- Non-residential investment OK
- Consumption held up well.

The Collapse, 2008-2009

From: Q3/2008 To: Q2/2009

Variable	Growth	×Share	=Contribution	Share of Growth
C	-2.0	67.2	-1.3	37.1
I	-24.3	16.1	-3.9	108.8
Residential	*-23.2*	*3.3*	*-0.8*	*21.3*
Non − Res	*-24.5*	*12.8*	*-3.2*	*87.6*
G	2.6	20.3	0.5	-14.9
NX	-30.5	-3.5	1.1	-30.0
Y	-3.6	100.0	-3.6	100.0

- After Lehman bankruptcy.
- Consumption fell but less than output.
- Consumption share of collapse was samll.
- Big driver of fall was non-res investment.

A previous bust, 1981-1982

From: Q3/1981 To: Q4/1982

Variable	Growth	×Share	=Contribution	Share of Growth
C	2.9	61.0	1.8	-69.3
I	-20.7	14.8	-3.0	119.7
Residential	*-12.0*	*4.5*	*-0.5*	*21.0*
Non − Res	*-24.4*	*10.3*	*-2.5*	*98.6*
G	4.0	24.3	1.0	-37.8
NX	-2005.0	0.0	-0.6	23.6
Y	-2.5	100.0	-2.5	100.0

- Investment has historically accounted for recessions.
- Consumption actually *grew* in the 1981-1982 recession.
- Worst since Depression...

The Recovery, 2009-2012

From: Q2/2009 To: Q2/2012

Variable	Growth	×Share	=Contribution	Share of Growth
C	6.0	68.3	4.1	58.6
I	36.7	12.7	4.7	66.4
Residential	*13.1*	*2.6*	*0.3*	*4.9*
Non − Res	*42.9*	*10.1*	*4.3*	*61.5*
G	-4.7	21.6	-1.0	-14.6
NX	23.6	-2.6	-0.6	-8.6
Y	7.0	100.0	7.0	100.0

- Consumption accounts for exactly its normal share.
- Strong investment...
- Offsets decline in G.

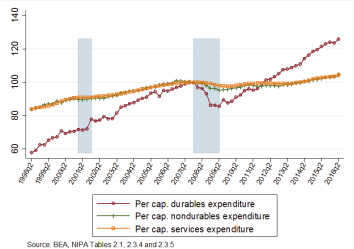
The Expansion, 2009-Present

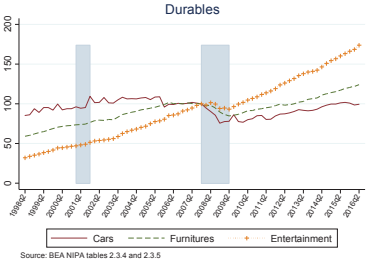
From: Q2/2009 To: Q4/2015

Variable	Growth	×Share	=Contribution	Share of Growth
C	15.5	68.3	10.6	72.7
I	56.1	12.7	7.1	49.0
Residential	*44.6*	*2.6*	*1.2*	*8.1*
Non − Res	*59.1*	*10.1*	*5.9*	*40.9*
G	-7.2	21.6	-1.5	-10.6
NX	54.7	-2.6	-1.4	-9.6
Y	14.5	100.0	14.5	100.0

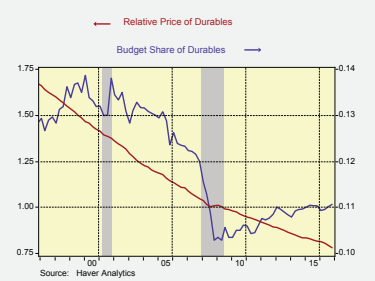
- Consumption accounts for exactly its normal share.
- Strong investment...
- Offsets decline in G.

Durables Consumption



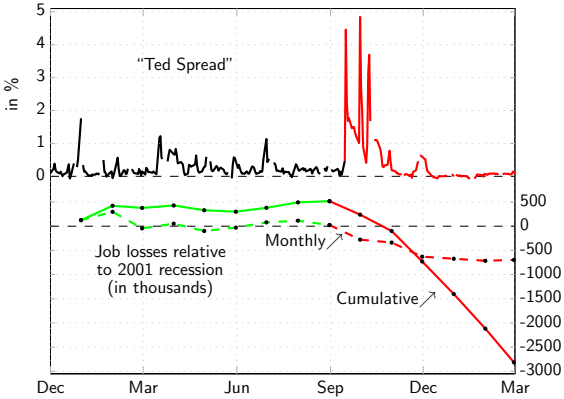


- Pistaferri (2016)
  First, during the late 1990s and early 2000s the growth in durables spending was sustained. This is what Hall (2011) calls the durables "buying frenzy" of the housing boom period. Durables are also more likely to be purchased with debt, and the easing of credit standards may have stimulated their purchase.

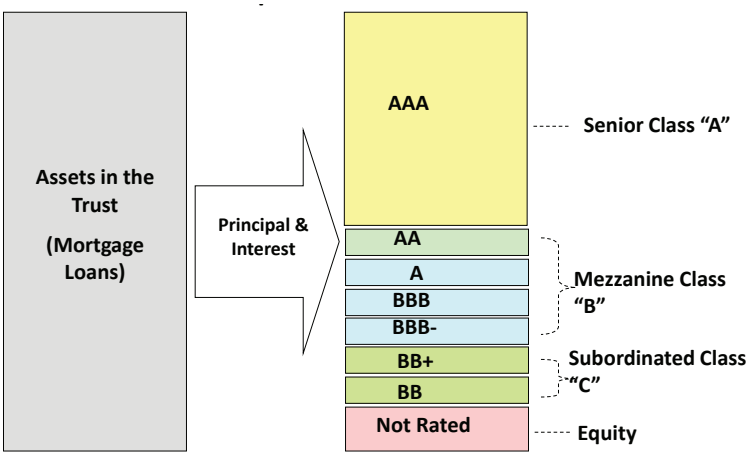


Timing of the Great Recession

- "Great" Recession started in December 2007
  - Ted Spread = LIBOR - Fed Funds rate
  - Financial markets were "OK"
  - Job losses mild
- Lehman fails on September 15, 2008
  - Worst job losses since Great Depression
- The big fall in *everything* happened after Lehman
  - Retail sales
  - Car sales
  - Investment
- *EXCEPT* Residential Investment.



Mortgages and Losses



- Investors lost $400 billion on AAA-rated securities tied to mortgages.

Ospina and Uhlig (2016)

- Three types of Deals

1: Prime Jumbo
  - loans too big for Fannie and Freddie
  - $\approx$ 1% subord. – 99% AAA

2: "Alt-A"
  - "A" credit
  - Typically reduced documentation
  - $\approx$ 5% sub. – 95% AAA

3: Subprime
  - Bad credit history
  - $\approx$ 20% sub. – 80% AAA

	Prime			Alt-A			Subprime		
Rating	**Before 2003**	**2003 - 2005**	**2006 - 2008**	**Before 2003**	**2003 - 2005**	**2006 - 2008**	**Before 2003**	**2003 - 2005**	**2006 - 2008**
AAA	0.0000	0.0032***	0.0259***	0.0002	0.0076***	0.0953***	0.0004	0.0013	0.0068***
	(0.0001)	(0.0008)	(0.0015)	(0.0003)	(0.0015)	(0.0015)	(0.0008)	(0.0018)	(0.0020)
AA	0.0001	0.2242***	0.5841***	0.0045**	0.2097***	0.7824***	0.0007	0.0348***	0.6260***
	(0.0007)	(0.0052)	(0.0093)	(0.0019)	(0.0060)	(0.0080)	(0.0040)	(0.0061)	(0.0061)
A	0.0006	0.2837***	0.3071***	0.0080***	0.3620***	0.7260***	0.0151***	0.1566***	0.7600***
	(0.0011)	(0.0075)	(0.0117)	(0.0027)	(0.0094)	(0.0119)	(0.0050)	(0.0077)	(0.0078)
BBB	0.0039***	0.3065***	0.2957***	0.0267***	0.4728***	0.5161***	0.0718***	0.3609***	0.8654***
	(0.0013)	(0.0088)	(0.0146)	(0.0033)	(0.0115)	(0.0117)	(0.0058)	(0.0091)	(0.0094)
BB	0.0163***	0.2850***	0.2303***	0.0499***	0.6415***	0.3546***	0.1113***	0.5755***	0.8861***
	(0.0021)	(0.0095)	(0.0143)	(0.0054)	(0.0173)	(0.0144)	(0.0199)	(0.0191)	(0.0166)
B	0.0336***	0.7159***	0.8828***	0.0863***	0.7765***	0.4816***	0.2448***	0.5133***	0.7463***
	(0.0026)	(0.0165)	(0.0344)	(0.0075)	(0.0254)	(0.0193)	(0.0419)	(0.0580)	(0.0495)
CCC	-	0.2474**	0.9484***	0.9710***	0.6840***	0.8850**	-	0.3836**	0.9931***
	(0.0754)	(0.1091)	(0.2731)	(0.0866)	(0.1954)	(0.3523)	(0.0977)	(0.1949)	(0.1512)
CC	-	0.0109	-	-	-	0.6322***	-	0.1189	-
	(0.1109)	(0.0951)	(0.2628)	(0.1027)	(0.0977)	(0.1078)	(0.1027)	(0.3823)	(0.5566)
C or Below	-	0.7679***	0.9687	0.4963***	-	0.3112***	-	0.3775	0.9932**
	(0.1215)	(0.1359)	(0.5928)	(0.0449)	(0.0219)	(0.0241)	(0.1369)	(0.3712)	(0.5020)
Observations	4,095	13,366	8,015	2,908	8,226	16,001	1,363	6,028	11,314
R-squared	0.0554	0.3468	0.4182	0.1571	0.4329	0.4975	0.1432	0.3217	0.7052
Weighted	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses
$*p < 0.10, **p < 0.05, ***p < 0.01$

Ospina and Uhlig (2016), Slide 2

Panel B: Losses by Mortgage Type and Credit Rating as of December 2013

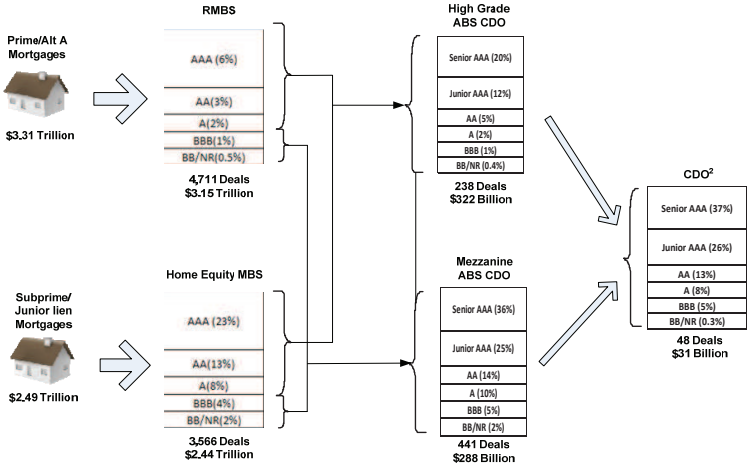
	Number of Securities			Dollar Amount		
	Total	W/ Losses	Pct. Losses	Total	W/ Losses	Pct. Losses
All Securities						
Prime	25,476.	11,987	47.1	1,238.7	37.5	3.0
Alt-A	27,135	17,763	65.5	1,327.3	145.2	10.9
Subprime	18,705	9,172	49.0	1,196.0	119.1	10.0
AAA Rated Securities						
Prime	15,610	5,418	34.7	1,172.7	14.8	1.3
Alt-A	14,851	7,646	51.5	1,210.0	78.9	6.5
Subprime	6,509	560	8.6	979.5	4.3	0.4
Investment Grade Ex-AAA Securities						
Prime	6,436	3,827	115.8	54.0	18.4	34.0
Alt-A	9,610	7,769	80.8	96.5	55.3	57.3
Subprime	10,893	7,456	68.4	203.9	104.8	51.4
Non-Investment Grade Securities						
Prime	3,430	2,742	79.9	12.0	4.3	36.2
Alt-A	2,674	2,348	87.8	20.8	10.9	52.7
Subprime	1,303	1,156	88.7	12.7	10.0	78.7

- Total losses: 301.2b
- Total AAA loss

Prime	14.8b
Alt-A	78.9b
Subprime	4.3b
Total	98b

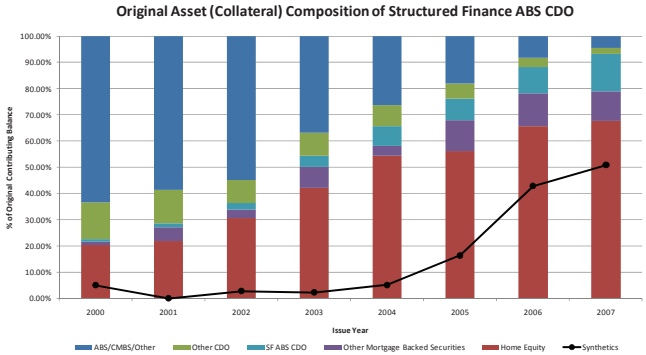
- $4.3b in losses on subprime
  - Subprime crisis?
- Where are the other $322b in losses in AAA losses?

Cordell, Huang, and Williams (2011)



- Collateralized Debt Obligations
- Originally
  - Collection of corporate bond

ABS CDOs



Note: This table summarizes shares of assets for structured finance ABS CDOs by five asset categories in the bar charts, and also includes shares of synthetic collateral in the line plot. Synthetics = Credit Default Swaps, Home Equity = Home Equity Mortgage-Backed Securities, SF ABS CDO = structured finance asset-backed security collateralized debt obiligations, ABS = asset-backed security, CMBS = commercial mortgage-backed securities, Other = all remaining asset collateral within the SF ABS CDO

- ABS CDOs were collections of Asset Backed Securities
  - Credit card receivables
  - Car loans
  - Appleby's Franchise revenue
  - Subprime ABS
- By 2007, mostly "HEL" = Subprime.
- Plus other CDOs...

Synthetics

SF ABS CDO Cash and Synthetic Collateral by Vintage 1999-2007				
Year	**Cash**	**Synthetic**	**Total**	**% Synthetic**
1999	304	-	304	0%
2000	6,391	600	6,991	9%
2001	14,891	-	14,891	0%
2002	13,456	3,000	16,456	18%
2003	25,431	-	25,431	0%
2004	52,327	6,186	58,513	11%
2005H1	25,808	3,827	29,636	13%
2005H2	65,071	15,346	80,416	19%
2006H1	52,608	18,501	71,109	26%
2006H2	83,287	77,315	160,602	48%
2007H1	73,948	49,065	123,013	40%
2007H2	26,230	27,516	53,746	51%
Total	**439,751**	**201,356**	**641,107**	**31%**

- Credit Default Swap:
  - Buyer of protection pays a premium
  - Seller of protection pays par if the bond defaults
- Arbitrage (assuming no other risks)

$$\text{Premium} = R_b - R_f$$

- Can construct a "synthetic bond"
  - Buy Riskless asset
  - Issue a CDS on the bond
- Synthetic Bond
  - Receive: $R_f + \text{premium} = R_b$
  - Worth nothing if default.

Making bonds?

Home Equity Bonds Placed into CDOs by Original Rating						
1998-2007						
Original Tranche Rating	Total # Issued	Total Placed in or Referenced in CDOs	% Placed or Referenced in CDOs	# Occurrences in CDOs	Share of Occurrences to Total Issued	Share of Occurrences to Total Placed in CDOs
AAA	12,948	1,463	11%	2,509	19%	171%
AA	5,486	3,893	71%	11,292	206%	290%
A	5,891	4,579	78%	14,714	250%	321%
BBB	6,993	5,496	79%	36,901	528%	671%
BB/B	1,976	899	45%	2,091	106%	233%
Amounts ($ Millions)						
Original Tranche Rating	Total $ Issued	Total Placed in or Referenced in CDOs	% Placed or Referenced in CDOs	$ Occurrences in CDOs	Share of $ Occurrences to Total Issued	Share of $ Occurrences to Total Placed in CDOs
AAA	2,011,451	160,090	8%	25,897	1%	16%
AA	173,150	125,152	72%	80,449	46%	64%
A	100,183	82,916	83%	71,245	71%	86%
BBB	77,025	64,061	83%	140,046	182%	219%
BB/B	18,002	9,055	50%	7,981	44%	88%

- For every one BBB bond, created 5 synthetic bonds
- For every dollar of subprime BBB, nearly tw dollars of potential losses.
- Most of this is now AAA.

The Boom

BBB-Rated Home Equity Bonds Placed into SF ABS CDOs by Issuance Year						
1998-2007						
Amounts ($ Millions)						
Time Period	BBB HE Bonds Issued	BBB HE Bonds Placed or Referenced in CDOs	% BBB Bonds Placed or Referenced in CDOs	BBB Bond Occurrences in CDOs	Share of $ of BBB HE Occurrences to Total Issued	Share of $ of BBB HE Occurrences to Total Placed in CDOs
1998	2,249	1,209	54%	603	27%	50%
1999	1,494	727	49%	435	29%	60%
2000	1,283	904	70%	792	62%	88%
2001	2,213	1,842	83%	1,708	77%	93%
2002	4,426	3,876	88%	3,443	78%	89%
2003	6,941	6,100	88%	6,985	101%	115%
2004	13,300	12,266	92%	21,898	165%	179%
2005H1	8,301	7,664	92%	25,254	304%	330%
2005H2	10,300	9,250	90%	29,934	291%	324%
2006H1	10,086	9,164	91%	34,060	338%	372%
2006H2	8,031	6,922	86%	12,124	151%	175%
2007H1	6,613	3,973	60%	2,706	41%	68%
2007H2	1,789	165	9%	104	6%	63%
Totals	77,025	64,061	83%	140,046	182%	219%

Notes: This table breaks out the dollar balance of Home Equity (HE) originally BBB-rated securities that were either placed or referenced in

- At the peak of the boom
  - For every dollar of BBB-rated Subprime
  - Banks created $3 of CDO paper
  - Most of it AAA
- Gigantic Side bet

CDO^2

SF ABS CDO Bonds Placed into CDOs by Original Rating 1998-2007						
Original Tranche Rating	Total # Issued	Total Placed in or Referenced in CDOs	% Placed or Referenced in CDOs	# Occurrences in CDOs	Share of Occurrences to Total Issued	Share of Occurrences to Total Placed in CDOs
Sr AAA	746	72	9.65%	134	17.96%	186.11%
Jr AAA	1,055	539	51.09%	1,572	149.00%	291.65%
AA	802	566	70.57%	2,544	317.21%	449.47%
A	**669**	**416**	**62.18%**	**2,402**	**359.04%**	**577.40%**
BBB	876	473	54.00%	1,739	198.52%	367.65%
BB/B	970	70	7.22%	81	8.35%	115.71%
Amounts ($ Millions)						
Original Tranche Rating	Total $ Issued	Total Placed in or Referenced in CDOs	% Placed or Referenced in CDOs	$ Occurrences in CDOs	Share of $ Occurrences to Total Issued	Share of $ Occurrences to Total Placed in CDOs
Sr AAA	$397,688	$26,934	6.77%	$1,490	0.37%	5.53%
Jr AAA	$137,852	$54,176	39.30%	$18,915	13.72%	34.91%
AA	$36,234	$26,837	74.07%	$23,387	64.55%	87.15%
A	**$18,656**	**$11,881**	**63.68%**	**$21,193**	**113.60%**	**178.38%**
BBB	$17,769	$10,251	57.69%	$9,731	54.76%	94.93%
BB/B	$19,017	$838	4.41%	$506	2.66%	60.41%

- $550 billion of AAA paper
- Lower rated stuff
  - Sold into other CDOs
  - Lehman sold to Merrill
  - Merrill sold to Lehman

Losses by year

Summary Expected Losses for All SF ABS CDOs by Issuance Year				
Vintage	**# Deals**	**Deal Balance ($ million)**	**Deal Loss ($ million)**	**Total Loss %**
1999	1	304	60	20%
2000	19	6,991	1,847	26%
2001	34	14,891	3,132	21%
2002	37	17,456	3,604	21%
2003	45	25,561	7,481	29%
2004	81	58,558	25,822	44%
2005	124	108,877	61,627	57%
2006	223	231,711	167,402	72%
2007	163	176,759	148,836	84%
Grand Total	**727**	**641,107**	**419,812**	**65%**
Notes: This table summarizes expected losses on all active SF ABS CDOs (as of March 2011) by vintage. SF ABS CDOs = Structured Finance CDOs.				
Source: Intex, Bloomberg, RBS (2011)				

- $550 billion of AAA paper
- Lower rated stuff
  - Sold into other CDOs
  - Lehman sold to Merrill
  - Merrill sold to Lehman

Losses by rating

Table 12 (a)

Distribution of Write-downs for SF ABS Tranches

Seniority	N Obs	Mean	Std Dev	5th Percentile	10th Percentile	25th Percentile	50th Percentile	75th Percentile	90th Percentile
Senior AAA	768	55%	35%	0%	0%	22%	59%	90%	100%
Junior AAA	1072	80%	35%	0%	1%	79%	100%	100%	100%
AA	803	91%	26%	0%	78%	99%	100%	100%	100%
A	669	93%	24%	0%	88%	100%	100%	100%	100%
BBB	876	91%	23%	1%	78%	96%	100%	100%	100%
BB and B	261	95%	16%	76%	91%	99%	100%	100%	100%
Not Rated	716	96%	18%	100%	100%	100%	100%	100%	100%

Note: This table summarizes the distribution of tranche losses by original rating or seniority.
Sources: Intex, Bloomberg, RBS(2011)

Table 12 (b)

Distribution of Write-downs for SF ABS Tranches by Vintage

Issue Year	N Obs	Senior AAA	Junior AAA	AA	A	BBB	BB and B	Not Rated
1999-2000	101	4%	10%	88%	66%	93%	67%	82%
2001	168	6%	9%	64%	11%	74%	81%	91%
2002	209	13%	20%	64%	65%	78%	91%	78%
2003	264	16%	44%	61%	63%	64%	93%	93%
2004	495	26%	71%	84%	95%	84%	99%	97%
2005	858	44%	83%	94%	94%	93%	97%	99%
2006	1696	67%	94%	99%	97%	96%	97%	99%
2007	1374	76%	93%	98%	99%	98%	96%	99%

Note: This table summarizes the distribution of tranche write-downs by original rating/seniority and issue year.
Sources: Intex, Bloomberg, RBS(2011)

- $550 billion of AAA paper
- Lower rated stuff
  - Sold into other CDOs
  - Lehman sold to Merrill
  - Merrill sold to Lehman

Losses by firm

Subprime and CDO Dealer Issuance Volumes				
Dealers	Dealer Shelf HE Issuance ($MM)	SF ABS CDO Issuance ($MM)	Senior AAA Issuance ($MM)	Credit Flux Writedowns ($MM)*
Merrill Lynch	$76,747	91,767	64,324	26,100
Citigroup	$54,032	70,552	50,945	34,106
UBS Warburg	$20,024	65,409	35,126	21,870
Goldman, Sachs & Co.	$52,727	59,103	45,313	-
Credit Suisse First Boston	$99,081	38,209	18,221	3,427
Deutsche Bank Securities	$59,635	37,362	16,363	2,092
Wachovia Securities	$12,528	25,033	24,363	1,860
RBS Greenwich Capital	$31,643	23,869	15,535	3,609
Calyon Securities	$0	22,825	14,399	3,400
Banc of America Securities	$28,524	22,617	15,911	9,089
Barclays Capital	$33,812	21,424	19,883	3,254
Lehman Brothers	$170,342	19,116	11,765	200
Bear Stearns	$61,698	19,111	9,985	2,300
Morgan Stanley	$121,289	14,237	7,356	7,800
WestLB Securities	$0	12,125	9,121	-
SG Corporate & Investment Bankii	$4,796	11,609	3,740	3,561
Dresdner Kleinwort Wasserstein	$0	9,875	7,087	1,604
JPMorgan	$53,628	7,584	4,081	1,300
Totals	$880,507	$571,826	$373,516	$125,572

- $550 billion of AAA paper
- Lower rated stuff
  - Sold into other CDOs
  - Lehman sold to Merrill
  - Merrill sold to Lehman

What went wrong?

- Recap:

Losses on Subprime AAA	\$4.3b
Losses on AAA SF ABS CDOS	$\sim$ \$300b

- most of collateral for latter was subprime.
- What went right with subprime?
- What went wrong with CDOs?

Fact 8: Investors understood the risks

- *Brookings Papers on Economic Activity*, Fall 2008: 69–145. Joint with Gerardi, Lehnert and Sherlund.
- Lehman Brothers: "U.S. ABS Weekly Outlook," August 15, 2005.
- "HEL Bond Profile Across HPA Scenarios"

#	Name	Scenario	Loss	Probability
(1)	Aggressive	11% HPA over the life of the pool	1.4%	15%
(2)		8% HPA for life	3.2%	15%
(3)	Base	HPA slows to 5% by end-2005	5.6%	50%
(4)	Pessimistic	0% HPA for the next 3 years 5% thereafter	11.1%	15%
(5)	Meltdown	-5% for the next 3 years, 5% thereafter	17.1%	5%

- Actual HPA: -10% annualized from Q4, 2005 to Q4, 2008
- Forecast losses as of 2/2009 in 2006-1 ABX from JPM: 23.44% (assuming -30% HPA in 2009!)

Coval, Jurek, and Stafford (2009a)

- Three bonds
  - $P(1) = 0.5$
  - $P(0) = 0.5$.
- What is $P(1, 1, 0)$?
- Portfolio of three bonds
  - E(payoff) $= 1.5$
- Issue three bonds
  1. AAA: 1 if payoff> 0
  2. AA: 1 if payoff> 1
  3. BBB: 1 if payoff> 2

			$\rho = 0$				$\rho = 1$			
#	Defaults	Payoff	p	AAA	AA	BBB	p	AAA	AA	BBB
1	0	3	1/8	1	1	1	1/2	1	1	1
2	1	2	3/8	1	1	0	0	1	1	0
3	2	1	3/8	1	0	0	0	1	0	0
4	3	0	1/8	0	0	0	1/2	0	0	0
P(default)				1/8	1/2	7/8		1/2	1/2	1/2

		Final rating					
	Initial rating	*Default correlation* (ρ)			*Default probability* (pD)		
	(ρ = 20%, pD = 5%)	40%	60%	80%	7.50%	10%	12.50%
CDO							
Junior	NR	D	C	CC	NR	NR	NR
Mezzanine	BBB−	BB−	B+	B+	B+	CCC	CC
Senior	AAA	A+	BBB−	BB	AAA	A+	BBB−
CDO^2 ([6, 12])							
Junior	C	D	NR	NR	NR	NR	NR
Mezzanine	AAA	B+	C	CC	BBB−	NR	NR
Senior	AAA	AAA	AAA	AA+	AAA	AAA	B−

Coval, Jurek, and Stafford (2009b)

- assume
  - log utility
  - Period 0 income=96/45
  - Period 1 income=bonds+1
- Argue in the paper that the excess returns are *too low* given the state prices.

#	Defaults	Payoff	p	AAA	AA	BBB	Output	State Price	q
1	0	3	1/8	1	1	1	4	$1/4 \times 1/8 = 1/32$	3/45
2	1	2	3/8	1	1	0	3	$1/3 \times 3/8 = 3/24$	12/45
3	2	1	3/8	1	0	0	2	$1/2 \times 3/8 = 3/16$	18/45
4	3	0	1/8	0	0	0	1	$1 \times 1/8 = 1/8$	12/45
P(default)				1/8	1/2	7/8			
Q(default)				12/45	30/45	42/45			
Excess Return				0.19	0.51	0.87			

Estimated correlations

- Correlation *was* assumed to be close to zero
  - Lots of BBB-rated bonds
  - 10-year default 3.13%
  - $0.03^9 0 = 1/\infty$.
- In 2011, correlation: 0.7.

Results: Loss Correlations by SF Asset Type Combinations
(with 95% confidence intervals)

Asset	Mfd. Hsg	Bank Cards	Auto Loans	RMBS
Mfd. Hsg.	**0.55** (0.518, 0.568)			
Bank Cards	**0.22** (0.208, 0.233)	**0.17** (0.162, 0.179)		
Auto Loans	**0.37** (0.356, 0.390)	**0.21** (0.201, 0.219)	**0.48** (0.464, 0.492)	
RMBS	**0.13** (0.124, 0.133)	**0.07** (0.069, 0.076)	**0.18** (0.172, 0.185)	**0.06** (0.061, 0.064)

EXHIBIT 6

S&P Proposed Correlations between Different ABS Asset Types in the Same Country, 2011

	CDO corp.	CDO SF	CDO other	ABS cons.	ABS com.	CMBS	RMBS	U.S. Agency	Guar.	FFELP
CDO corp.	0.70	0.30	0.30	0.30	0.30	0.30	0.30	0.20	0.15	0.10
CDO SF	0.30	0.80	0.30	0.45	0.45	0.45	0.45	0.30	0.20	0.20
CDO other	0.30	0.30	0.70	0.30	0.30	0.30	0.30	0.20	0.15	0.20
ABS consumer	0.30	0.45	0.30	0.70	0.40	0.40	0.40	0.20	0.15	0.20
ABS commercial	0.30	0.45	0.30	0.40	0.70	0.40	0.40	0.20	0.15	0.20
CMBS	0.30	0.45	0.30	0.40	0.40	0.70	0.40	0.20	0.15	0.20
RMBS	0.30	0.45	0.30	0.40	0.40	0.40	0.70	0.30	0.15	0.20
U.S. Agency	0.20	0.30	0.20	0.20	0.20	0.20	0.30	0.70	0.20	0.30
Guaranteed	0.15	0.20	0.15	0.15	0.15	0.15	0.15	0.20	0.70	0.20
FFELP	0.10	0.20	0.20	0.20	0.20	0.20	0.20	0.30	0.20	0.70

Source: Ghetti et al. [2011, Table 4].

Why did they get the correlation wrong?

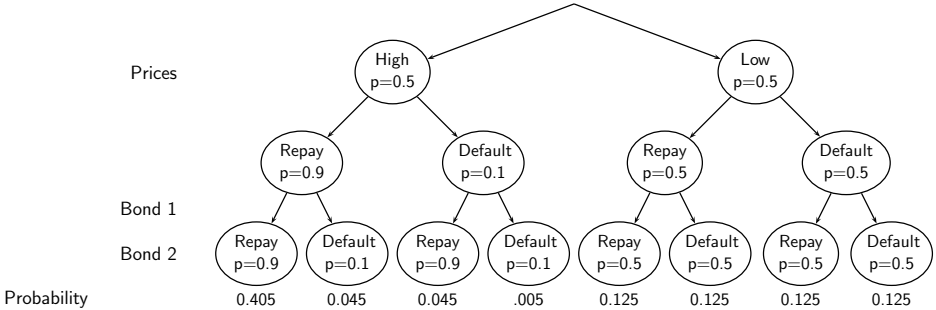
- According to Coval, Jurek, and Stafford (2009a)
  ... since substantial lending to subprime borrowers is a recent phenomenon, historical data on defaults and delinquencies in this sector of the mortgage market is scarce. The possibility for errors in the assessment of default correlations, the default probabilities and the ensuing recovery rate for the securities was significant.
- My theory in July of 2008 was similar
  - Subprime had never been stressed with declining house prices.
- Then in August, I figured out that I was wrong.
- If CJS were right, why were there only 4.3b losses on AAA subprime?

Understanding the problem

- Conditional on price
  - $\text{corr}(x_1, x_2) = 0$
  - mortgages are uncorrelated.
- In a period of stable prices
  - Observed correlation will be low
- Unconditional

$$\text{corr}(x_1, x_2) = \frac{P_{DD} - P_D^2}{P_D - P_D^2} = 0.04/.21 = 0.19$$

- Correlation driven by macro variable



Why the difference?

- True model

$$\begin{aligned} y_1 &= \beta x + \varepsilon_1 \\ y_2 &= \beta x + \varepsilon_2 \end{aligned}$$

- Assume $\text{var}(x) = \sigma_x^2$ and $\text{var}(\varepsilon_1) = \text{var}(\varepsilon_2) = \sigma_\varepsilon^2$, $\text{cov}(\varepsilon_1, \varepsilon_2) = 0$

Mortgage Modeling

- Estimate β
- Put in a scenario for house prices
- Who did it this way?
  - Rating agencies when analyzing mortgages
  - Lehman
  - "Mortgage people"

Bond modeling

- Estimate

$$\begin{aligned} \text{corr}(y_1, y_2) &= \frac{\text{cov}(\beta x+\varepsilon_1, \beta x+\varepsilon_2)}{\text{std}(\beta x+\varepsilon_1)\text{std}(\beta x+\varepsilon_2)} \\ &= \frac{\beta^2\sigma_x^2+\text{cov}(\varepsilon_1,\varepsilon_2)}{\sigma_x^2+\sigma_\varepsilon^2} \\ &= \frac{\beta^2\sigma_x^2}{\sigma_x^2+\sigma_\varepsilon^2} \end{aligned}$$

- $\sigma_x^2 \to 0 \Rightarrow \text{corr} \to 0$, $\sigma_x^2 \to \infty \Rightarrow \text{corr} \to 1$
- Who did this way?
  - Rating agencies when analyzing CDOs
  - "Corporate finance people"

Fact 1: Federal Subsidies to Banks Were Small

- Capital Purchase Program
- Government bought $435 billion of preferred stock from banks
- How much did this cost?
- Two popular approaches:
  1. Government gave $435 billion to banks.
  2. Banks paid the money back with interest – government made money!
- Both are problematic.
- Truth is somewhere in-between

Did the government give money to banks?

- No.
- The government invested in the banks.
- Think of the government owning a portfolio of bonds
  - Future tax revenues.
- In the fall of 2008, government sold bonds and invested the cash in preferred stock of banks
- If government paid the market price, has no effect on government net worth.
  - Cost is zero
- Did the government pay the market price?
  - No.
- The subsidy is the difference between the what the government paid and the market price

Estimates of the subsidy

- Example: PNC Bank
  - On October 24, 2008, Treasury bought $7.6 billion of PNC preferred stock.
  - On February 5, 2010, PNC paid back the money plus $421m in interest plus a gain of $320m on warrant sales in October of 2010.
  - Overall Treasury made a $740 million profit on investment in PNC
- BUT, that was just one possible outcome – unknown in 2008
- What was known is that the Treasury was taking on risk
- TARP Congressional Oversight Panel said

Amount	Value	Subsidy	Subsidy Rate
$7.6b	$5.5b	$2.1b	27%

Estimates of cost of TARP CPP

- Different *ex ante* estimates of the cost of the CPP

Date	Analysis	Amount	Value	Subsidy	Subsidy Rate
12/2008	CBO	\$243b	\$182	\$61	25%%
2/2009	COP	\$254b	\$176	\$78	31%%

- CBO also estimated that the Fed policies to aid market cost another \$21 billion
- Total cost of assistance to markets: $\leq$\$100 billion.
- Household equity fell by trillions of dollars.
  - \$100 billion of relief would have made little difference

The slide you've all been waiting for...

- The end.

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